Transferring CNN With Adaptive Learning for Remote Sensing Scene Classification

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***Abstract*— Accurate classification of remote sensing (RS) images is a perennial topic of interest in the RS community. Recently, transfer learning, especially for fine-tuning pretrained convolutional neural networks (CNNs), has been proposed as a feasible strategy for RS scene classification. However, because the target domain (i.e., the RS images) and the source domain (e.g., ImageNet) are quite different, simply using the model pretrained on an ImageNet dataset presents some difficulties. The RS images and the pretrained models need to be properly adjusted to build a better classification system. In this study, an adaptive learning strategy for transferring a CNN-based model is proposed. First, an adaptive transform is used to adjust the original size of the RS image to a certain size, which is tailored to the input of the subsequent pretrained model. Then, an adaptive transferring model is proposed to automatically learn what knowledge from the pretrained model should be transferred to the RS scene classification model. Finally, in combination with a label smoothing approach, an adaptive label is presented to generate soft labels based on the statistics of the classification model predictions for each category, which is beneficial for learning the relationships between the target and nontarget categories of scenes. In general, the proposed methods adaptively manage the input, model, and label simultaneously, which leads to better classification performance for RS scene classification. The proposed methods are tested on three widely used datasets, and the obtained results show that the proposed methods provide competitive classification accuracy compared to the state-of-the-art methods.**

***Index Terms*— Convolutional neural network (CNN), remote sensing (RS) scene classification, transfer learning.**

I. Introduction

With the advancements of Earth observation technology, the resolution of remote sensing (RS) images has constantly improved over the past few decades [1]. These images tend to contain plentiful and complex structural information about the Earth's surface, which presents tremendous challenges to further interpretation. As a crucial problem for understanding the RS imagery, scene classification is an area of active interest to the RS community. It aims to assign a predefined semantic class to a query RS image, as illustrated in Fig. 1. In recent years, numerous studies on RS scene classification have been conducted, actuated by practical applications, such as land resource management [2], urban planning [3], environment monitoring [4], [5], and geospatial object detection [6], [7].

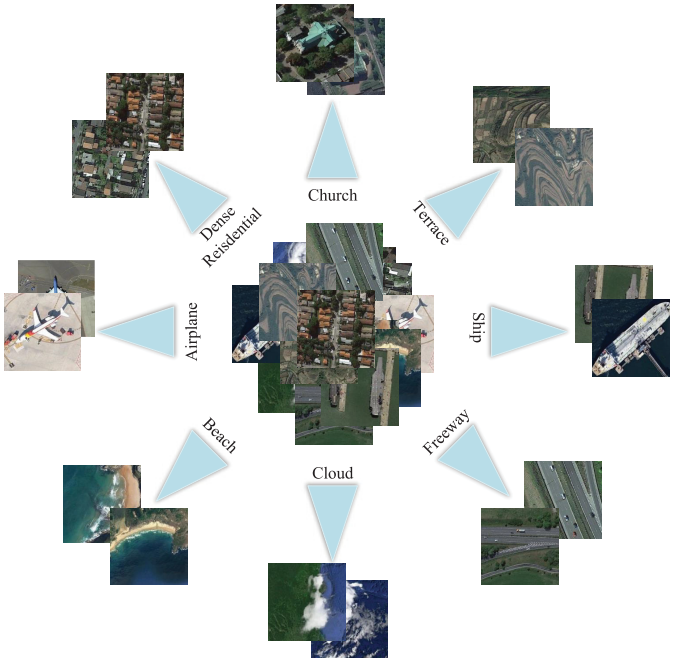


Fig. 1. Illustration of the content of RS scene classification.

For the past several decades, methods of RS scene classification have made vast progress. Early studies for RS scene classification were transplanted from strategies designed for natural images, which led to unsatisfying performance due to the distinctive differences between natural images and RS images. To remedy this limitation, various models that represent RS images with discriminative features have been proposed sustainably [8]–[10]. However, these methods still could not capture the most discriminative features between different categories of RS images. Fortunately, deep learning (DL) theory has developed rapidly in recent years [11]. The classical models based on convolutional neural network (CNN), such as AlexNet [12], VGGNet [13], GoogLeNet [14], and ResNet [15], stand out among numerous DL-based methods because of their vigorous feature extraction capability.

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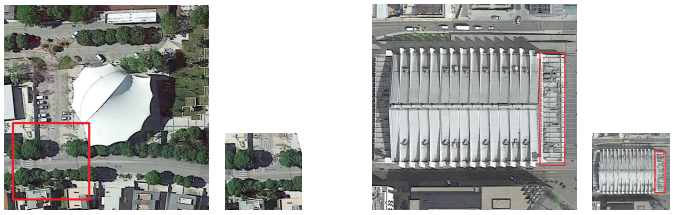
These CNN-based models are widely applied in computer vision tasks, including RS scene classification [16], [17].

Nevertheless, unlike natural image classification, the number of labeled samples for RS scene classification is usually limited. Training a deep CNN model with such limited data faces enormous obstacles, and a common issue is the poor generalization capability of the trained model. A popular method to mitigate this problem is the transfer learning strategy, which exploits the limited RS images to fine-tune a CNN-based model (e.g., AlexNet, VGGNet, or ResNet) that has been pretrained on a large-scale dataset (e.g., ImageNet [18], Openimage, and Places365). The researchers in the RS scene classification field attempted to use pretrained CNN models as feature extractors to generate information with high semantic discrimination. Hu *et al.* [19] presented two strategies for extracting CNN features with respect to different layers. The research attempted to form better representations for RS scenes from CNN activations. For further improving the performance of land-use classification, Zhao *et al.* [20] proposed a transfer learning method with fully pretrained CNNs. In this method, a multilayer perceptron classifier was pretrained with the high-level features extracted by the feature descriptor of the pretrained CNNs. Deng *et al.* [21] designed a framework combining deep CNN with context features, which transferred the pretrained CNN as a feature extractor to acquire convolutional features. These transfer learning-based methods tend to construct an end-to-end learning structure and achieve satisfying performance in the RS scene classification task.

Although fine-tuning CNN-based models provide an effective strategy for RS scene classification, such approaches still have several limitations in view of the characteristics of RS images. These can be summarized from three perspectives, related to input, model, and label, which are discussed in the following.

First, with respect to the inputs of pretrained CNN models, due to the matrix transformation conducted in the fully connected layers, the input RS images must be processed into a certain fixed size, which is the same as that processed in the source dataset (e.g., 224 $\times$ 224 pixels in ResNet pretrained on the ImageNet dataset). However, RS images always have larger sizes than the required size of pretrained CNN models (e.g., the Aerial Image dataset [22] has a size of 600 $\times$ 600). The most common solutions for this issue randomly crop or interpolate the original image [23]–[25], as shown in Fig. 2. The former operation might lose the class-specific ground object [center in Fig. 2(a)], while the latter inevitably discards detailed information from the original RS image [the fringe pattern in Fig. 2(b)]. Either of them is detrimental to the performance of RS scene classification.

Second, with reference to the model, the majority of fine-tuning strategies applied between heterogenous networks commonly require manual configurations of layer associations or feature selections, depending on the source task (e.g., classification of the ImageNet dataset) and the RS scene classification task [26], [27], which requires an exhaustive tuning procedure for the anticipated performance and might not be optimal. In other words, since different layers of CNN generate various levels of representations, the information extracted from a pretrained CNN-based model has varying levels of relevance in an RS scene classification model. From the perspective of knowledge transferring, some source information is more important than others, while some are irrelevant or even harmful depending on the task difference. For example, since networks generate different levels of representations at the different levels of abstractions [13], the information of lower layers might be more useful when the input domains of the tasks are similar. This is because higher level features are more task-specific, while lower level features are more task-agnostic. Yet, this is not a common issue when the source and target tasks are discrepant. There is no doubt that natural images and RS image scenes are semantically distant, and current transfer learning-based methods have no mechanism to identify which source information is vital to transfer for RS scene classification.



(a) (b)

Fig. 2. Results of different preprocessing methods for original RS scenes. (a) Random cropping. (b) Interpolation.

Third, with respect to the label, most of the pretrained CNN models used in RS scene classification are trained by hard labels, in which 1 is assigned to the target category and 0 to the nontarget categories [28]–[30]. In fact, adopting such treatment is inappropriate for RS scene classification due to the highly complex spatial patterns and geometric structures in RS images. These scenes always have larger intraclass diversity and interclass similarity [31], [32]. For example, two subclasses of Fig. 1 depict a "Church" scene and a "Dense Residential" scene. Both scenes contain buildings, trees, and roads. The differences between them are embodied in architectural styles, space arrangements, and the density of the buildings. Meanwhile, in Fig. 2(a), the class-specific ground object is the center, but other categories, such as residential, parking plot, and trees (similar to forest), still exist. Hence, assigning identity values of labels to such categories is apparently unreasonable without considering the real relationships among various scenes. One tempting approach to changing labels is label smoothing [33], which is an effective regularization tool for easing overfitting problems caused by deep CNNs. It uses a series of soft labels by taking an average of the hard labels and the uniform distribution over labels. However, it still treats the nontarget categories as fixed identity probability. To exploit the information between nontarget categories, a way to adjust the values of labels effectively is still needed.

To tackle these issues, in this article, an adaptive learning strategy for transferring CNN called T-CNN is proposed for the purpose of RS scene classification. The proposed method comprises three main methods: the adaptive

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transform (T-CNN-AT), the adaptive model (T-CNN-AM), and the adaptive label (T-CNN-AL). First, T-CNN-AT adaptively transforms the original RS images into a certain size for the input of a pretrained CNN-based model. Then, T-CNN-AM is utilized to autonomously decide what information to transfer between the source and RS scene classification model; this is the main structure of T-CNN. Finally, T-CNN-AL combines label smoothing with the model predictions to consecutively update the labels during the training procedure. By combining T-CNN-AT, T-CNN-AM, and T-CNN-AL into the unified framework, the behavior of the RS scene classification can be intrinsically enhanced.

The key contributions of this article are fourfold.

1) To address the difficulties that current learning-based approaches have in transferring knowledge between heterogenous CNN-based models, an effective model called T-CNN-AM, which is based on metanetworks, is proposed for transfer learning. The proposed approach automatically decides which information from the pretrained model is relevant for RS scene classification and which layers should be transferred from the pretrained model to the RS scene classification model.
2) In order to solve the problem of mismatching between the original size of RS images and the input size of pretrained CNN models, based on spatial transformer network (STN) [34], T-CNN-AT is designed to adaptively transform the original RS images to the size required by pretrained CNN models. T-CNN-AT reserves the original information of the RS images as much as possible and is capable of learning the effective transformations of various RS images that are beneficial for RS scene classification.
3) Given the genuine relationships among different categories of RS scenes, T-CNN-AL, which combines label smoothing and the predictions of an RS scene classification model, is proposed to constantly update labels at each training epoch; the updated labels are used to supervise the RS scene classification model until convergence takes place. T-CNN-AL leverages the regularization of label smoothing and the statistics of model predictions to better learn the relationships between the target categories and nontarget categories of RS scenes.
4) Combining T-CNN-AT, T-CNN-AM, and T-CNN-AL, an end-to-end framework called T-CNN is proposed for RS scene classification. T-CNN works in a totally adaptive way in terms of preprocessing, knowledge transference, and training procedures. Experimental results show that the proposed T-CNN outperforms a number of state-of-the-art methods on three benchmark datasets, and the adaptive modules are able to combine with a state-of-the-art method.

The remainder of this article is organized as follows. Section II reviews selected related works. Section III introduces the proposed T-CNN in detail. Section IV exhibits the results from experiments conducted on three publicly available datasets. Finally, the conclusion of this article is presented in Section V.

II. Related Works

A. RS Scene Classification

RS scene classification is a fundamental task in the RS community, and plenty of studies have been conducted for scene classification due to its wide applications. Scene-level RS scene classification aims to classify each given RS image patch into a semantic category. An essential step of this procedure is determining how to represent visual data with discriminative features.

Early feature extraction methods of RS scene classification mainly relied on human-engineering descriptors, such as scale-invariant feature transformation [35] and histograms of oriented gradients [36]. Both methods utilized locally handcrafted features to learn regional information, resulting in discrimination of the global features. An improved strategy is encoding these local descriptors through the use of encoding approaches (e.g., the improved Fisher kernel [37], vector of locally aggregated descriptors [38], and bag-of-visual-words [39]). These methods have been proved to be effective tools for RS scene classification although such handcrafted features do have limitations in terms of representation capability. Given this circumstance, unsupervised feature learning methods have been developed to learn features from unlabeled images automatically, an approach that has become a competitive alternative to handcrafted feature models [9], [10], [40]. However, these unsupervised methods still cannot capture the most discriminative features between different categories.

Fortunately, thanks to the advancements of the DL theory and the increased capacity of high-performance computing with parallel resources, DL-based algorithms have recently sprung up in the RS scene classification community [41]–[43]. CNN-based methods stand out among the numerous DL-based methods due to their vigorous feature extraction capability. Li *et al.* [44] proposed a key region method aiming at retaining the global information of RS images, a method that could be combined with various CNN models to improve the performance of RS scene classification. Wang *et al.* [45] conducted an enhanced feature pyramid network with deep semantic embedding for classification. It improved on vanilla CNNs by extracting multiscale and multilevel features within the whole RS image. In [46], a novel paradigm to design a suitable CNN architecture to be used in continuous search space with the gradient-based optimization method was investigated. Experiments on various datasets proved the effectiveness of the neural architecture search strategy. Comparing these CNN-based methods with the handcrafted feature models and unsupervised methods, it can be concluded that CNN-based methods can learn more class-specific and generalized features automatically while only needing original input RS images without extra feature extractors, which enhances the accuracy of RS scene classification.

B. Transfer Learning

Although numerous CNN-based models have achieved remarkable performance in RS scene classification, the amount of annotated data for RS scene classification is rather limited in comparison to natural image classification. Training a deep

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CNN model with such limited data presents enormous obstacles: the usual issue is the poor generalization capability of this model. To remedy this limitation, a considerable number of efforts have been made to improve the performance of CNN, including data augmentation [47], analyzing adversarial perturbations [48], and transferring the pretrained CNN model into an RS scene classification task [19].

Transfer learning plays a crucial role in current CNN-based classification models. In fine-tuning, the model for RS scene classification utilizes parameters trained on other large datasets, such as ImageNet. Many attempts at RS scene classification have obtained favorable results. Nogueira *et al.* [27] presented the performance of CNNs for RS scene classification with different fine-tuning strategies. Li *et al.* [49] proposed a method using transfer learning and the deep CNN model with limited RS scene samples, which was demonstrated by sufficient experiments. In [50], the cross-sensor RS scene understanding problem was formulated as a heterogeneous network-oriented transfer learning problem, and the knowledge between heterogeneous networks was transferred using the pseudolabel recursive propagation mechanism. Generally, transfer learning takes advantage of the pretrained convolutional layers and fully connected layers to adapt the structure to the new tasks, a technique that has achieved great success in the RS scene classification community.

III. Methodology

In this section, the details of the proposed methods are illustrated, including model architecture and specific algorithms. First, the integrated framework of the proposed T-CNN for RS scene classification is introduced. Then, the three primary methods of the T-CNN-adaptive transform (T-CNN-AT), the adaptive model (T-CNN-AM), and the adaptive label (T-CNN-AL) are elaborated successively.

A. Overview of the Proposed Strategy

Let the RS image dataset be $\mathcal{H} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C represent the height, the width, and the channels of the RS images, respectively. Suppose that the dataset $\mathcal{H}$ contains N labeled images $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N\} \in \mathbb{R}^{H\times W\times C}$ and the corresponding set of labels $\boldsymbol{Y} = \{y_1, y_2, \ldots, y_N\} \in \mathbb{R}^{\mathcal{K}}$, where $\mathcal{K}$ is the number of categories.

Fig. 3 shows the framework of the proposed model. First, all raw RS images are preprocessed by T-CNN-AT, which transforms the input images automatically. The output size of the processed RS images is tailored to the input size of the pretrained CNN model and the RS scene classification model. Then, the processed RS images are fed into T-CNN-AM. During this phase, T-CNN-AM autonomously decides which feature maps of the pretrained model are relevant for RS scene classification and which layers should be transferred from the pretrained model to the RS scene classification model. Through metanetworks, the weights of different layers between the pretrained model and RS scene classification model are required to measure the amount of knowledge being transferred. Finally, T-CNN-AL leverages the predictions of the RS scene classification model to constantly update labels at each training epoch, and the updated labels are used to supervise the RS scene classification model until the model is convergent.

B. Adaptive Transform

The size of RS images always varies with different scenes and datasets. However, many available pretrained CNN models often need input images to be resized into a fixed size to generate input vectors of the size required by fully connected layers. These models often adopt interpolation or random cropping to solve this issue directly, which undoubtedly discards much information in the scenes and, thus, deteriorates the classification performance. Moreover, RS images are categorized into an independent set of semantic-level land use and land cover categories based on image contents. As a matter of fact, categories of RS images depend on regions containing class-specific ground objects, e.g., airplane, bridge, tennis court, and others. These objects in different RS images tend to have similar outlines but have different views and scales. Based on the learnable module of the STN, improved T-CNN-AT can automatically learn an effective way of transforming original RS images into a certain form without losing key information, which is conducive to improving the performance of RS scene classification.

As shown in Fig. 4, T-CNN-AT is split into three parts: generate a network, grid template, and mapper. First, generate network takes an RS image as input and, through a number of hidden layers, outputs a parametric matrix that is used to affine transformation. Then, the predicted transformation parameters should be applied to generate a sampling grid, which comprises a set of points. The grid template is used to sample these points from the input RS images and then produce the transformed output. Finally, the input RS images and grid template are fed to the mapper to produce the transformed RS image, which is sampled from the input RS image at the grid template. The details of these three parts are given as follows.

Let $\boldsymbol{x}_i$ be an input RS image of $\boldsymbol{X}$ for the generate network. Through the generate network, the parametric matrix $\boldsymbol{\Theta}$ can be formulated as

$$\boldsymbol{\Theta} = f_{\text{gen}}(\boldsymbol{x}_i) \tag{1}$$

where $f_{\text{gen}}(\cdot)$ denotes the generate network function containing convolutional layers and fully connected layers.

T-CNN-AT is aimed at warping the input RS image $I \in \mathbb{R}^{H\times W\times C}$ into a new form, and each output pixel is calculated by applying a mapping kernel at a particular location in the input RS image. For clarity of exposition, the output pixels are defined to lie on a grid template $\mathcal{G} = \{G_j\}$ of pixels $G_j = (x_j^{\text{target}}, y_j^{\text{target}})$ to form an output image $O \in \mathbb{R}^{H'\times W'\times C}$, where H' and W' are the height and width of the grid, respectively. H' and W' are equal to the input size of the subsequent pretrained CNN model. As different RS datasets always possess various sizes of images, T-CNN-AT is able to control the size of the grid by a scale factor $\varepsilon = (H'W'/HW)^{1/2}$. The values of ε vary with different RS datasets.

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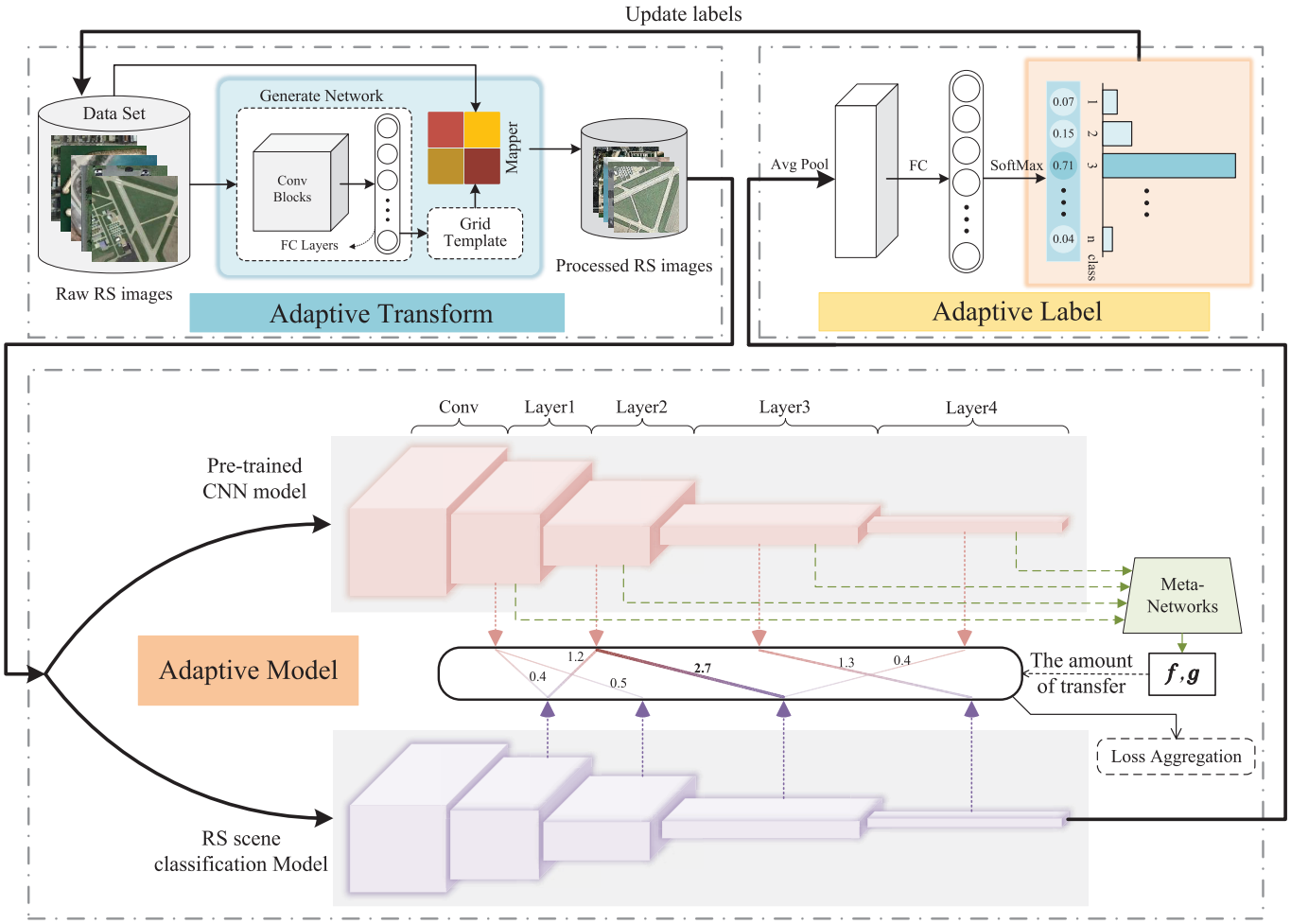


Fig. 3. Framework of the proposed T-CNN for RS scene classification.

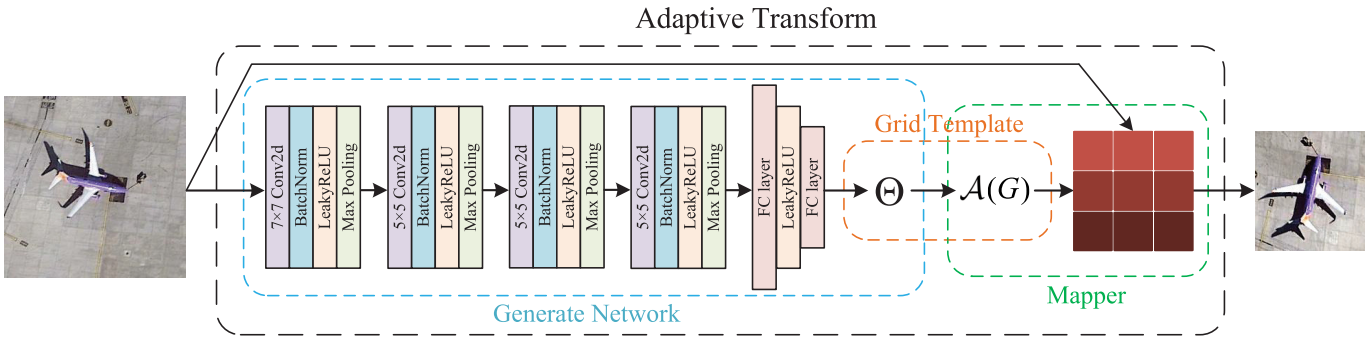


Fig. 4. Illustration of T-CNN-AT.

Assume that $\mathcal{A}$ is a 2-D affine transformation whose parameters are placed by $\boldsymbol{\Theta}$, and the pointwise transformation is denoted as

$$\begin{pmatrix} x_j^{\text{source}} \\ y_j^{\text{source}} \end{pmatrix} = \mathcal{A}(G_j) = \boldsymbol{\Theta} \begin{pmatrix} x_j^{\text{target}} \\ y_j^{\text{target}} \\ 1 \end{pmatrix} = \begin{bmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \end{bmatrix} \begin{pmatrix} x_j^{\text{target}} \\ y_j^{\text{target}} \\ 1 \end{pmatrix}$$

$$= \begin{bmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{bmatrix} \begin{pmatrix} x_j^{\text{source}} \\ y_j^{\text{source}} \end{pmatrix} + \begin{bmatrix} \theta_{13} \\ \theta_{23} \end{bmatrix} \quad (2)$$

where $(x_j^{\text{target}}, y_j^{\text{target}})$ are the target coordinates of the grid template in the output image, and $(x_j^{\text{source}}, y_j^{\text{source}})$ are the source coordinates in the input RS image corresponding to the mapping points. In addition, the height and width coordinates are normalized to $[-1, 1]$. According to (2), T-CNN-AT requires only six parameters of $\boldsymbol{\Theta}$, where θ_{11}, θ_{12}, θ_{21}, and θ_{22} are able to rotate and scale the input image, and θ_{13} and

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θ_{23} are capable of translating the image. It is worth noting that T-CNN-AT predicts geometric information from the generated network, but the warp parameters of $\boldsymbol{\Theta}$ keep the track of and pass through the whole network, which is different from the original STN, which feeds the warped images into the subsequent network. One of the major drawbacks of feeding the warped image directly is that the output image samples only from the cropped input image; pixel information outside the cropped region is discarded, which may discard the background information of RS images. T-CNN-AT ameliorates this drawback because pixel information outside the transformed area is also preserved until the last transformation.

The parametric matrix $\boldsymbol{\Theta}$ probably contains decimals, which makes the indexes of the pixels in the input image also decimals. Consequently, (3) denotes a method that samples the indexes of points into integers. The whole process is called mapper, which takes the set of mapping points $\mathcal{A}(\mathcal{G})$, along with the input RS image, and creates the output image O. Each $(x_j^{\text{source}}, y_j^{\text{source}})$ in $\mathcal{A}(\mathcal{G})$ denotes the spatial location in the input where a mapping kernel is applied to get the value at a particular pixel in the output O. It can be described as

$$O_j^c = \sum_m^H \sum_n^W I_{mn}^c \kappa\left(x_j^{\text{source}} - m; \Omega_x\right) \bullet \kappa\left(y_j^{\text{source}} - n; \Omega_y\right) \tag{3}$$

where $\forall j \in \{1, \ldots, H'W'\}, \forall c \in \{1, \ldots, C\}$, I_{mn}^c is the value at location (m, n) in channel c of the input I, and O_j^c is the output value for pixel j at location $(x_j^{\text{target}}, y_j^{\text{target}})$ in channel c. The image interpolation is denoted by Ω_x and Ω_y, which represents a common sampling kernel $\kappa(\cdot)$. In this work, a bilinear sampling kernel is used.

Through this mapping mechanism, T-CNN-AT allows normal backpropagation of the loss, which is helpful when located at the beginning of the pretrained network and the RS scene classification model, leading to an end-to-end model for the RS scene classification task.

C. Adaptive Model

Transfer learning with pretrained CNN has been successfully applied to RS scene classification. The commonly used transfer models usually transfer knowledge from the pretrained CNN model to the RS scene classification model (e.g., a heterogenous CNN model) by requiring layer association or feature selection manually. Such methods have no mechanism to identify which information from the source domain it is vital to transfer for the RS scene classification task. To this end, T-CNN-AM is proposed to learn which knowledge of the pretrained CNN network is significant for RS scene classification and which layers should be transferred from the pretrained CNN model to the RS scene classification model.

In order to decide which information is relevant for RS scene classification, the loss should be formalized to force this effect. For the RS scene classification task, $\boldsymbol{x}_i \in \mathbb{R}^{H' \times W' \times C}$ and y_i are processed RS image (processed by T-CNN-AT) and its class label. Let $\mathcal{S}^a(\boldsymbol{x}_i)$ be intermediate feature maps of the ath layer of the pretrained CNN network $\mathcal{S}$. The goal is to train the RS scene classification model $\mathcal{T}_\vartheta$ with parameter ϑ using the knowledge of $\mathcal{S}$. The expression $\mathcal{T}_\vartheta^b(\boldsymbol{x}_i)$ denotes intermediate feature maps of the bth layer of the RS scene classification model. To transfer the knowledge from $\mathcal{S}^a(\boldsymbol{x}_i)$ to $\mathcal{T}_\vartheta^b(\boldsymbol{x}_i)$, the ℓ_2 objective is described as follows:

$$\left\| t_\vartheta\left(\mathcal{T}_\vartheta^b(\boldsymbol{x}_i)\right) - \mathcal{S}^a(\boldsymbol{x}_i) \right\|_2^2 \tag{4}$$

where t_ϑ represents a linear transformation parameterized by ϑ (e.g., a pointwise convolution). This method can be called feature comparison (FC).

For most transfer learning-based methods, the source model and the target model are usually trained for different tasks. Certainly, not all the intermediate features of the pretrained CNN model are beneficial to learning the RS scene classification task. Hence, an FC loss, which can emphasize the channels based on their utility for the RS scene classification task, is designed as follows:

$$\mathcal{L}_{\text{FC}}^{a,b}\left(\vartheta \,|\boldsymbol{x}_i, \eta^{a,b}\right) = \frac{1}{HW} \sum_c \eta_c^{a,b} \sum_{u,v} \left(t_\vartheta\left(\mathcal{T}_\vartheta^b(\boldsymbol{x}_i)\right)_{c,u,v} - \mathcal{S}^a(\boldsymbol{x}_i)_{c,u,v} \right)^2 \tag{5}$$

where $u \in \{1, \ldots, H\}, v \in \{1, \ldots, W\}$, and $\eta_c^{a,b}$ is the nonnegative weight of channel c satisfying $\sum_c \eta_c^{a,b} = 1$. Channel weights are set as a function, $\eta^{a,b} = [\eta_c^{a,b}] = f_\varphi^{a,b}(\mathcal{S}^a(\boldsymbol{x}_i))$, which can be realized by taking the softmax output of a metanetwork. The metanetwork takes features of pretrained CNN models as the input, and φ represents the parameter of the metanetwork. $\eta^{a,b}$ operates the dot product over the L_2 distance of $t_\vartheta(\mathcal{T}_\vartheta^b(\boldsymbol{x}_i))$ and $\mathcal{S}^a(\boldsymbol{x}_i)$. Due to the possible inconsistent spatial or spectral size of $\mathcal{S}^a(\boldsymbol{x}_i)$ and $\mathcal{T}_\vartheta^b(\boldsymbol{x}_i)$, bilinear interpolation and learnable convolutional layers play the role of linear transformation function t_ϑ in metanetwork to preprocess input feature maps so that $t_\vartheta(\mathcal{T}_\vartheta^b(\boldsymbol{x}_i))$ can do pointwise operation with $\mathcal{S}^a(\boldsymbol{x}_i)$.

Deciding the pair (a, b) is a crucial part of T-CNN-AM when the knowledge is transferred from the pretrained CNN model to the RS scene classification model. To achieve this purpose, a learnable parameter $\rho^{a,b} = g_\varphi^{a,b}(\mathcal{S}^a(\boldsymbol{x}_i)) \geq 0$ for each pair (a, b) is introduced, which is aimed at automatically deciding the amount of transfer between the ath and bth layers of the pretrained CNN and RS scene classification model. Thus, given the channel weights η and weights of compared

Algorithm 1 Learning of ϑ With Metaparameter φ

Input: Data set $\boldsymbol{X}_{train} = \{(\boldsymbol{x}_i, y_i)\}$, Learning rate μ, Total training epochs E;

```
for e = 0; e < E; e++ do
    Sample a batch X_b ⊂ X_train with |X_b| = B;
    minimize (1/B) Σ_{(x,y)∈X_b} L_Total(ϑ|x_i, y_i, φ);
    Initialize: ϑ_0 ← ϑ
    for t = 0; t < T − 1; t++ do
        ϑ_{t+1} ← ϑ_t − μ∇_ϑ (1/B) Σ_{(x,y)∈X_b} L_Transfer(ϑ|x_i, φ);
    end for
    ϑ_{T+1} ← ϑ_T − μ∇_ϑ (1/B) Σ_{(x,y)∈X_b} L_Origin(ϑ_T|x_i, y_i);
end for
```

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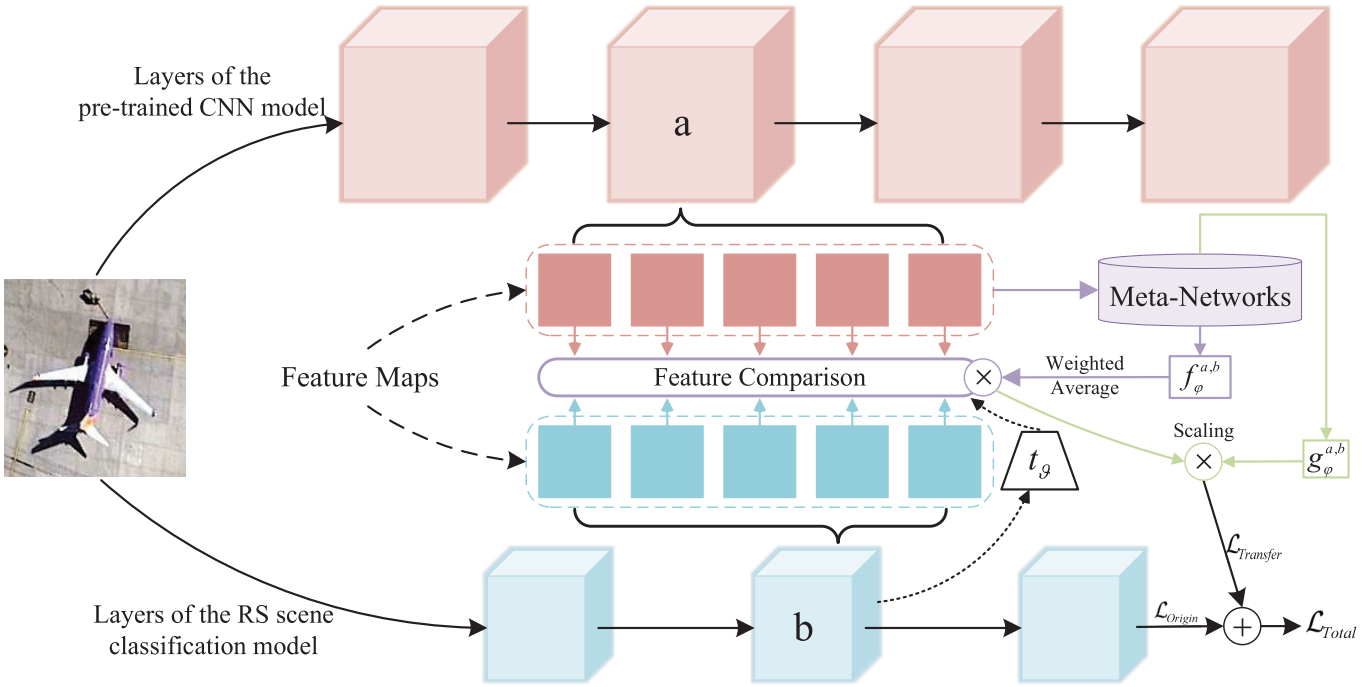


Fig. 5. Principle of T-CNN-AM.

pairs ρ, the combined transfer loss is denoted as

$$\mathcal{L}_{\text{Transfer}}(\vartheta|\boldsymbol{x}_i,\varphi)=\sum_{(a,b)\in\mathcal{C}}\rho^{a,b}\mathcal{L}_{\text{FC}}^{a,b}\left(\vartheta|\boldsymbol{x}_i,\eta^{a,b}\right) \tag{6}$$

where $\mathcal{C}$ represents a set of candidate pairs. To train the RS scene classification model, the final loss L_{total} is given as

$$\mathcal{L}_{\text{Total}}(\vartheta|\boldsymbol{x}_i,y_i,\varphi)=\mathcal{L}_{\text{Origin}}(\vartheta|\boldsymbol{x}_i,y_i)+\alpha\mathcal{L}_{\text{Transfer}}(\vartheta|\boldsymbol{x}_i,\varphi) \tag{7}$$

where $\mathcal{L}_{\text{Origin}}$ is the original loss (e.g., cross-entropy) and α is a hyperparameter that remains positive. Fig. 5 illustrates the principles of T-CNN-AM.

The goal of T-CNN-AM is to achieve high performance on the RS scene classification task when the RS scene classification model is learned using the $\mathcal{L}_{\text{Total}}(\vartheta|\boldsymbol{x}_i,y_i,\varphi)$. To this end, $\mathcal{L}_{\text{Transfer}}(\cdot|\boldsymbol{x}_i,\varphi)$ should encourage learning of useful features for the RS scene classification task. To increase the effectiveness of the FC decided by metanetworks' parameterized by φ, the proposed training scheme, which adopts the concepts of the bilevel scheme [51], is formally outlined in Algorithm 1.

To train the RS scene classification model jointly with metanetworks, the RS scene classification model parameters ϑ and the metanetwork parameters φ should be updated alternatively. First, update the RS scene classification model for a single step with $\mathcal{L}_{\text{Total}}(\vartheta|\boldsymbol{x}_i,y_i,\varphi)$. Then, metanetworks' parameters φ should be updated with the following steps.

1) Given the present RS scene classification model parameters $\vartheta_0=\vartheta$, update the RS scene classification model for T times by gradient-based algorithms for minimizing $\mathcal{L}_{\text{Transfer}}$. Thus, the resulting parameter ϑ_T is learned only with the knowledge of the pretrained CNN model.
2) Update ϑ to minimize $\mathcal{L}_{\text{Origin}}$ once. It is a one-step adaptation, which updates ϑ_T to ϑ_{T+1}. The resulting object $\mathcal{L}_{\text{Origin}}(\vartheta_{T+1})$ is capable of measuring how fast the RS scene classification model has adapted to the RS scene classification task.
3) Update metaparameter φ by minimizing $\mathcal{L}_{\text{Origin}}(\vartheta_{T+1})$.

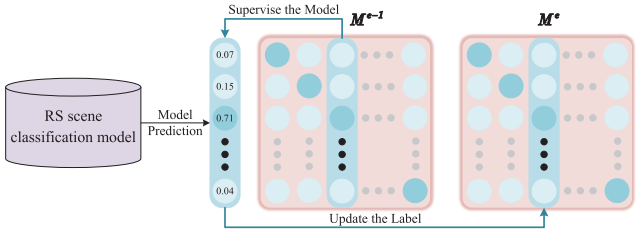


Fig. 6. Procedure of T-CNN-AL.

In addition, the weights of the pretrained CNN, which is used as the source model, are not updated in the training scheme. Although the parameters of the source model are not updated, they work together with metanetworks to compute channel weights. Then, the parameters of the RS scene classification model will be updated accordingly. Moreover, the weights of the RS scene classification model are initialized with the pretrained weights for accelerating the training process.

D. Adaptive Label

Most of the existing pretrained CNN models treat the labels of nontarget categories with identity values, which is inappropriate for RS scene classification due to the high degree of intraclass diversity and interclass similarity in RS images. Label smoothing, which uses soft labels by taking an average of the hard labels and the uniform distribution over labels, is aimed at providing regularization for a learnable

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classification model [33]. However, equally treating each nontarget category could weaken the capability of label smoothing and limit the model performance. Moreover, in [52], it has been proven that model predictive distributions provide a promising way to reveal the implicit relationships among different categories. Inspired by this knowledge, T-CNN-AL is proposed to generate more reliable soft labels that consider the relationships among different categories.

T-CNN-AL is used in the training process of the RS classification model. The training dataset $\boldsymbol{X}_{\text{train}} = \{(\boldsymbol{x}_i, y_i)\}$ contains $\mathcal{K}$ categories, where $\boldsymbol{x}_i \in \mathbb{R}^{H' \times W' \times C}$ has been processed by T-CNN-AT. For each sample $(\boldsymbol{x}_i, y_i)$, the RS scene classification model predicts the probability $p(k|\boldsymbol{x}_i)$ of the class k. The distribution q of the hard label satisfies $q(k = y_i|\boldsymbol{x}_i) = 1$ and $q(k \neq y_i|\boldsymbol{x}_i) = 0$. Label smoothing leverages a uniform distribution to soften the distribution of hard labels. Thus, the modified cross-entropy loss can be denoted as follows:

$$\mathcal{L}_{\text{LS}} = -\sum_{k=1}^{\mathcal{K}} \left((1-\delta) q(k|\boldsymbol{x}_i) + \frac{\delta}{\mathcal{K}} \right) \log p(k|\boldsymbol{x}_i) \tag{8}$$

where δ represents the smoothing parameter.

It is natural to think that the similarity among different categories can be required by the model predictions. As a consequence, T-CNN-AL utilizes model predictions from the RS scene classification model to dynamically update the labels, called adaptive labels, during the training procedure. Specifically, given a sample $(\boldsymbol{x}_i, y_i)$, if the prediction is correct, the adaptive label corresponding to y_i will be updated by $p(\boldsymbol{x}_i)$. Subsequently, the new adaptive labels will be used to supervise the model. The schematic of T-CNN-AL is illustrated in Fig. 6.

Let E be the number of the training epochs. Define $\mathcal{M} = \{M^0, M^1, \ldots, M^e, \ldots, M^{E-1}\}$ as a set of matrices of the classwise adaptive labels at different epochs, where $M^e \in \mathbb{R}^{\mathcal{K} \times \mathcal{K}}$, and each column of M^e corresponds to the soft labels for a certain class. Suppose that the current training epoch is e; the adaptive label $M^{e-1}_{y_i}$ represents the adaptive labels of the target category y_i. The training adaptive loss of the RS scene classification model supervised by $M^{e-1}_{y_i}$ is denoted as

$$\mathcal{L}_{\text{Adaptive}} = -\sum_{k=1}^{\mathcal{K}} M^{e-1}_{y_i,k} \log p(k|\boldsymbol{x}_i). \tag{9}$$

In practice, the model is difficult to converge because of the random parameter initialization and the lack of the correct information from the labels. The feasible strategy combines the adaptive labels with label smoothing labels, and the total training loss is represented by

$$\mathcal{L} = \beta \mathcal{L}_{\text{LS}} + (1-\beta) \mathcal{L}_{\text{Adaptive}} \tag{10}$$

where β is a hyperparameter to balance $\mathcal{L}_{\text{LS}}$ and $\mathcal{L}_{\text{Adaptive}}$. As for initialization, at the beginning of the eth training epoch, the adaptive label M^e is initialized as a zero matrix. Until the input sample $(\boldsymbol{x}_i, y_i)$ is classified correctly, the y_i column in M^e can be updated by predicted results $p(\boldsymbol{x}_i)$, namely, $M^e_{y_i,k} = M^e_{y_i,k} + p(k|\boldsymbol{x}_i)$. At the end of the eth training epoch, the cumulative M^e column by column is normalized as

$$M^e_{y_i,k} \leftarrow \frac{M^e_{y_i,k}}{\sum_{\ell=1}^{\mathcal{K}} M^e_{y_i,\ell}}. \tag{11}$$

In addition, at the first epoch, the uniform distribution is exploited to initialize each column in M^0.



Fig. 7. Several examples of scenes in the UCM dataset. The categories are (a) airplane, (b) beach, (c) dense residential, (d) harbor, and (e) tennis court.

IV. Experiments

In this section, sufficient experiments are implemented to evaluate the effectiveness of the proposed method. First, different benchmark datasets and evaluation metrics used in the experiments on RS scene classification are briefly described. Then, the implementation details of the experiments, for instance, structural design and parameters settings, are introduced. Finally, to demonstrate the advantages of the proposed method in this article, comparisons of classification performance on various benchmarks with state-of-the-art methods are displayed and analyzed.

A. Datasets

Three widely used datasets, the UCMerced Land-Use dataset (referred to as UCM) [39], the Ariel Image dataset (referred to as AID) [22], and the NWPU-RESISC45 dataset (referred to as NWPU-45) [2], are utilized in all experiments.

1) UCM: The UCM dataset is the first ground-truth dataset generated from RS images, which are all provided by the United States Geological Survey. This dataset has 2100 RS scene images containing three RGB channels, each of which is categorized into a certain category. The images in UCM contain 21 typical land-use categories, including agricultural, airplane, baseball diamond, beach, and others. Each category possesses 100 RS images measuring 256 × 256 with a 0.3-m resolution. Fig. 7 shows several examples of scenes in the UCM dataset.

2) AID: The AID dataset is a large-scale dataset for RS images that were published by the Huazhong University of Science and Technology and Wuhan University in 2017. It comprises 30 scene categories, each of which has about 220–420 aerial images, for a total of 10 000 aerial images, including airport, commercial, church, dense residential, farmland, parking, square storage tanks, and others. Each image is 600 × 600 pixels in RGB space with spatial resolution varying from about 8 to about 0.5 m. Some samples of the AID are shown in Fig. 8.

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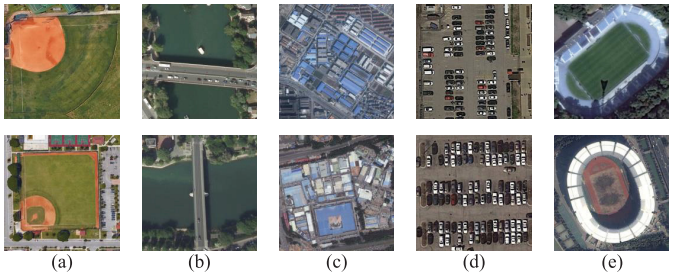


Fig. 8. Some examples of scenes in the AID dataset. The categories are (a) baseball field, (b) bridge, (c) industrial, (d) parking lot, and (e) stadium.

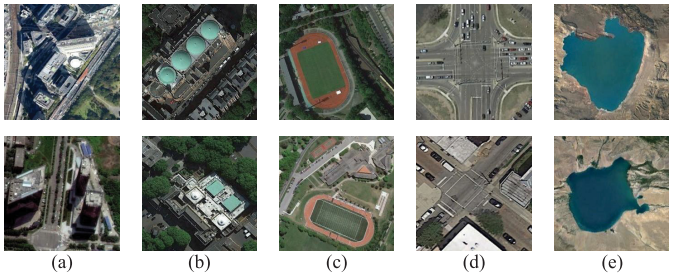


Fig. 9. Some samples of scenes in the NWPU-45 dataset. The categories are (a) commercial area, (b) church, (c) ground track field, (d) intersection, and (e) lake.

3) NWPU-45: The NWPU-45 dataset is one of the largest RS scene datasets in terms of the number and category of images, with greater intraclass differences and interclass similarities than the aforementioned two datasets. It contains 31 500 scene images with the size of 256 × 256 in RGB color space, which are divided into 45 scene categories, including basketball court, circular farmland, medium residential, mobile home park, parking lot, rectangular farmland, and others. The spatial resolution of the images ranges from about 30 to 0.2 m per pixel. Fig. 9 displays some examples of scenes in the NWPU-45 dataset.

B. Evaluation Metrics

Accuracy has great significance for metrics in RS image recognition tasks. Overall accuracy (OA), average accuracy (AA), and confusion matrix (CM) are three widely used evaluation metrics. These indices are illustrated as follows.

1) *Overall Accuracy:* OA is defined as the number of correctly classified images divided by the total number of images in each category during the testing procedure and reflects the overall classification performance of the network.
2) *Average Accuracy:* AA is the average value of accuracy for each category in test datasets.
3) *Confusion Matrix:* CM is an information table used to analyze errors and confusions between different categories. It is acquired by counting the correct and incorrect classifications of each type of test image while accumulating the results in the table. The CM is equipped to show detailed classification results. It focuses on misclassification and acts as a supplementary evaluation index for OA.

In this article, all experimental datasets have a balanced amount of each category. Under the circumstances, AA is equivalent to OA. As a consequence, OA and CM are adopted to evaluate the performance of different models.

TABLE I

OVERALL ACCURACIES AND STANDARD DEVIATIONS (%) OF THE PROPOSED MODEL AND COMPARISON MODELS ON THE UCM DATASET

Models	OA (80% Training Ratio)
VGG-VD-16 [22]	95.21 ± 1.20
CNN-FT-Full [20]	96.67 ± 0.40
VGG-S [30]	96.92
D-CNN [43]	98.93 ± 0.10
Inception-v3-CapsNet [53]	99.05 ± 0.24
SF-CNN [54]	99.05 ± 0.27
MG-CAP [55]	99.0 ± 0.10
GBN [56]	98.57 ± 0.48
CNN-MS2AP [57]	99.01 ± 0.42
DA2Net [58]	97.74 ± 0.22
T-CNN	**99.33 ± 0.11**

The entry with the highest values is bold-faced.

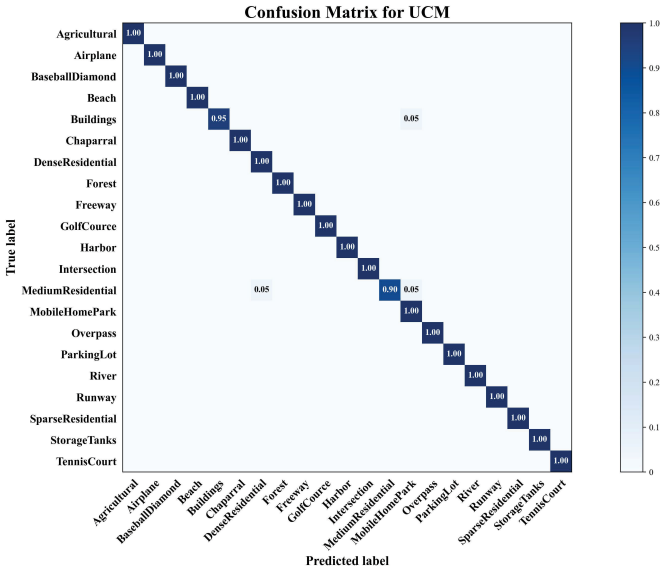


Fig. 10. CM of the proposed method on the UCM dataset.

C. Experimental Settings

1) Dataset Settings: The training set and the test set are divided randomly by conventional proportions on different datasets. For the UCM dataset, 80% of the images in each category are used to train, and the rest of the images are used to test. With regard to the AID dataset, two group experiments with 20% and 50% data are performed on the training procedure, and the rest are used as test data, respectively. As for the NWPU-45 dataset, the training ratio of the whole dataset is set to 10% and 20%, respectively. These training ratios

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TABLE II
OVERALL ACCURACIES AND STANDARD DEVIATIONS (%) OF THE PROPOSED MODEL AND COMPARISON MODELS ON THE AID DATASET

Methods	OA	
	20% Training Ratio	50% Training Ratio
VGG-VD-16 [22]	86.59 ± 0.29	89.64 ± 0.36
Inception-v3-CapsNet [53]	93.79 ± 0.13	96.32 ± 0.12
SF-CNN [54]	93.60 ± 0.12	96.66 ± 0.11
SCCov [59]	93.12 ± 0.25	96.10 ± 0.16
MG-CAP [55]	93.34 ± 0.18	96.12 ± 0.12
GBN [56]	92.20 ± 0.23	95.48 ± 0.12
ResNet50-EAM [60]	93.64 ± 0.25	96.62 ± 0.13
CNN-MS2AP [57]	92.19 ± 0.22	94.82 ± 0.20
ACNet [61]	93.33 ± 0.29	95.83 ± 0.29
DA2Net [58]	93.11 ± 0.52	–
ACR-MLFF [62]	92.73 ± 0.12	95.06 ± 0.33
T-CNN	**94.55 ± 0.27**	**96.72 ± 0.23**

The entry with the highest values is bold-faced.

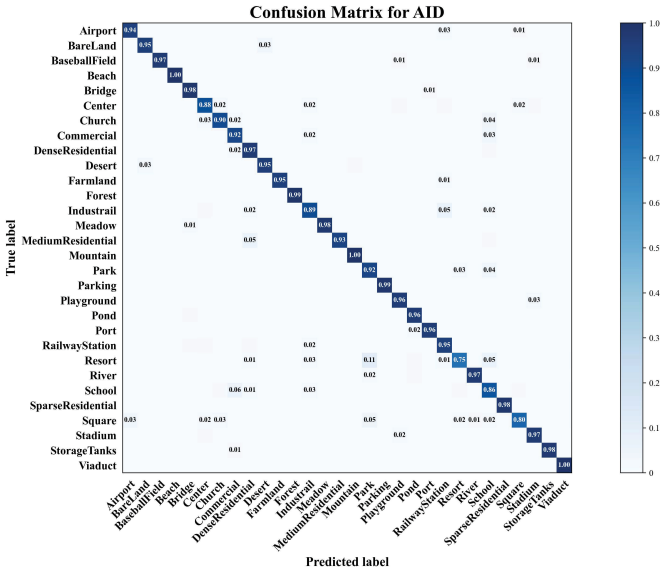


Fig. 11. CM of the proposed method on the AID dataset with 20% training data.

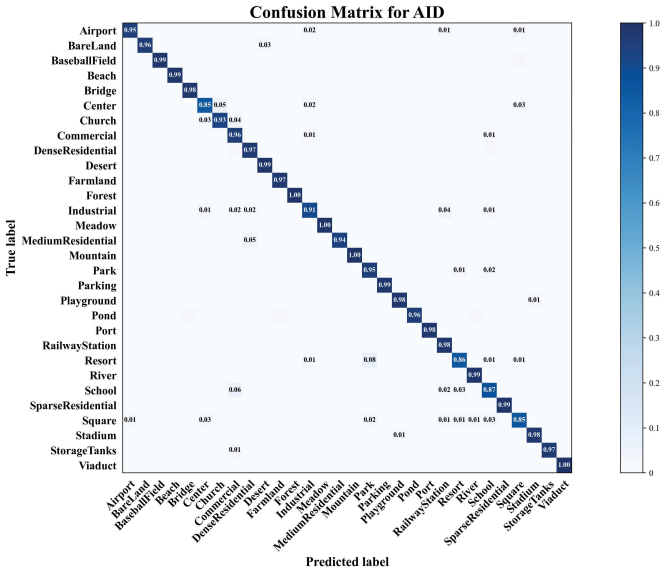


Fig. 12. CM of the proposed method on the AID dataset with 50% training data.

have been confirmed based on previous research into the RS scene classification task so that the proposed method can be compared with the latest methods.

In addition, several augment approaches, e.g., horizontal flipping, vertical flipping, and random rotation from $-90°$ to $90°$, are performed on the training data. In order to avoid one-time occasionally, for each aforementioned training ratio, the experiment is repeated five times in order to calculate the average and standard deviation of the accuracies.

2) Structural Parameter Settings: To evaluate the proposed T-CNN strategy, the ImageNet dataset is selected as the source data and the UCM, AID, and NWPU-45 datasets as the target data. For these datasets, 34- and 18-layer ResNets are used as the source and RS scene classification models, respectively. The pretrained weights of these models come from the model library of PyTorch [63]. Since the ResNet-18 is the backbone network to several RS image datasets, the last fully connected layer of the ResNet18 model should be replaced with a new fully connected layer, which is used to obtain the final classification results of the RS scene classification task.

In this article, metanetwork is used to transfer knowledge from the pretrained CNN model (ResNet-34) to the RS classification model (ResNet-18) adaptively. Convolutional layers of ResNet-34 and ResNet-18 are both divided into four blocks, so we can get 16 pairs in total. For each pair $(a, b) \in \mathcal{C}$, where $\mathcal{C}$ is the set of candidates of pairs, metanetwork is constructed as a one-layer fully connected network for processing feature maps. This structure takes the global average pooled features of the ath layer of the source network as an input, and η and ρ as outputs. In addition to the fully connected structure, softmax activation is used to generate the channel weights $\eta_c^{a,b}$ while satisfying $\sum_c \eta_c^{a,b} = 1$. Moreover, for the transfer amount ρ between layers, except for the fully connected structure, ReLU6 is selected to guarantee nonnegativeness of ρ and to prevent $\rho^{a,b}$ from becoming too large [64].

With regard to T-CNN-AT, it is placed before the RS scene classification model (e.g., ResNet-18) to transform the original images into a suitable size automatically, adapting to the input size of the particular RS scene classification model. The generate network of each T-CNN-AT module comprises four convolutional layers and pooling layers. The stride values of the convolutional layer and pooling layer are set to 1 and 2, respectively. Bach normalization layers [65] and LeakyReLU activations with a 0.1 negative slope coefficient are also used. Following the convolutional layers, two fully connected layers are performed to generate the parametric matrix $\mathbf{\Theta}$.

3) Training Settings: The stochastic gradient descent and the Nesterov momentum with a momentum weight of 0.9 are used to train all models. The training epochs are 50, and

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TABLE III

OVERALL ACCURACIES AND STANDARD DEVIATIONS (%) OF THE PROPOSED MODEL AND COMPARISON MODELS ON THE NWPU-45 DATASET

Methods	OA	
	10% Training Ratio	20% Training Ratio
Fine-tuned VGGNet-16 [2]	87.15 ± 0.45	90.36 ± 0.18
D-CNN [43]	89.22 ± 0.50	91.89 ± 0.22
Inception-v3-CapsNet [53]	89.03 ± 0.21	92.60 ± 0.11
SF-CNN [54]	89.89 ± 0.16	92.55 ± 0.14
RAN [66]	88.79 ± 0.53	91.40 ± 0.30
SCCov [59]	89.30 ± 0.35	92.10 ± 0.25
CNN-MS2AP [57]	87.91 ± 0.19	90.98 ± 0.21
GLANet [67]	89.50 ± 0.26	91.50 ± 0.17
MF2Net [68]	90.17 ± 0.25	92.73 ± 0.21
ACR-MLFF [62]	90.01 ± 0.33	92.45 ± 0.20
T-CNN	**90.25 ± 0.14**	**93.05 ± 0.12**

The entry with the highest values is bold-faced.

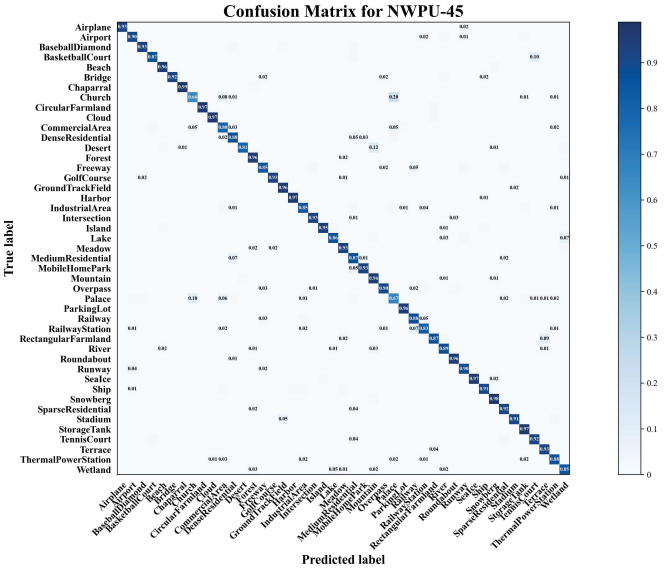


Fig. 13. CM of the proposed method on the NWPU-45 dataset with 10% training data.

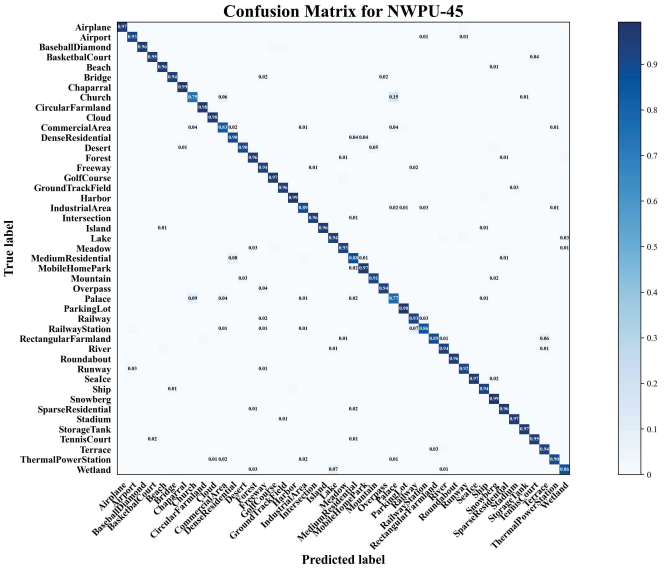


Fig. 14. CM of the proposed method on the NWPU-45 dataset with 20% training data.

the batch size is 16. We use a weight decay of 10^{-4} and an initial learning rate of 0.01 and decay the learning rate with a cosine annealing. During the FC procedure, α is set to 0.5. As for metanetworks, the Adam optimizer is used with a learning rate of 10^{-3} or 10^{-4} and a weight decay of 0 or 10^{-4}. In the metatraining scheme, $T = 2$ is enough to learn the transfer strategy. With regard to T-CNN-AL, β is set to 0.5, and the label smoothing parameter δ is set to 0.1. All experiments are carried out on a PC with CPU i7-10700F, 64 GB of RAM, 1T SSD, and GPU RTX-3070. The PyTorch open-source framework and Python programming language are used to implement the experiments.

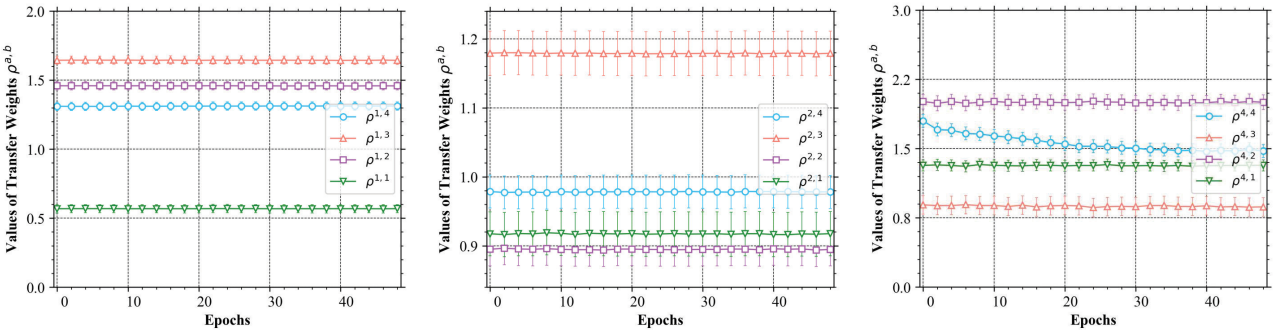
D. Results and Analysis

In this section, the performance of the proposed method on three benchmark datasets is displayed and analyzed. To validate the proposed method extensively, we compare it with several state-of-the-art RS scene classification models over different datasets. Moreover, some indispensable discussions are further illustrated appropriately to present the generality and robustness of the proposed model.

1) Results of the UCM Dataset: Owing to the limited number of RS images in each category, the task to classify various categories is challenging on the UCM dataset. We use 80% of the RS images for training and the rest for testing. The OA values and their standard deviations are summarized in Table I. It can be concluded that the performance of all the tested models is acceptable, while T-CNN achieves the best performance on the test dataset. Compared to the other methods, the OA values achieved by T-CNN outperform the other models by 4.12 (VGG-VD-16), 2.66 (CNN-FT-Full), 2.41 (VGG-S), 0.4 (D-CNN), 0.28 (Inception-v3-CapsNet and SF-CNN), 0.33 (MG-CAP), 0.76 (GBN), and 0.32 (CNN-MS2AP) percentage points. Except for the OA values, the meliority of T-CNN is also verified by CM, as shown in Fig. 10. Given the limited space, we only exhibit the CM generated by T-CNN over three benchmark datasets. It is obvious that the misclassification primarily appears between different types of architecture (i.e., buildings, mobile home park, and residential), which are similar in terms of RS scenes. These encouraging results demonstrate that it is feasible to classify different scenes within the UCM dataset by employing T-CNN.

2) Results of the AID Dataset: The AID dataset is much more intricate than the UCM dataset due to the interclass similarity. To study the behavior of T-CNN on the AID dataset deeply, two training datasets with different ratios of the whole data (e.g., 20% and 50%) are established. As shown in Table II, T-CNN obtains the best performance among all scene classification models in both cases. Particularly, in terms of

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(a) (b) (c)

Fig. 15. Mean values and standard deviations of transfer parameter $\rho^{a,b}$ between the layers of the pretrained CNN model and the RS scene classification model on three benchmark datasets. (a) $S^1(x)$. (b) $S^2(x)$. (c) $S^3(x)$.

OA, T-CNN obtains 94.55% and 96.72% in the condition of 20% and 50%, respectively. With 20% RS images used to train, the improvements achieved by T-CNN are 7.96 (VGG-VD-16), 0.76 (Inception-v3-CapsNet), 0.95 (SF-CNN), 1.43 (SCCov), 1.21 (MG-CAP), 2.35 (GBN), 0.91 (ResNet50-EAM), 2.36 (CNN-MS2AP), and 1.22 (ACNet) percentage points. When 50% RS images are utilized to train different models, the enhancements obtained by T-CNN are 7.08 (VGG-VD-16), 0.40 (Inception-v3-CapsNet), 0.06 (SF-CNN), 0.62 (SCCov), 0.60 (MG-CAP), 1.24 (GBN), 0.10 (ResNet50-EAM), 1.90 (CNN-MS2AP), and 0.89 (ACNet) percentage points. In addition to the OAs, the CMs of T-CNN confirmed by the AID dataset are displayed in Figs. 11 and 12. From the observation, it can be concluded that T-CNN is adept at classifying RS images, including forest, meadow, mountain, and viaduct. However, it is also apparent that the behavior of T-CNN on the AID dataset is not as good as that on the UCM dataset. One explanation is that the semantics of the AID dataset are more complicated than those of the UCM, resulting in more misclassifications of various scenes.

3) Results of the NWPU-45 Dataset: The NWPU-45 dataset has the largest scale of the three datasets. Hence, only 10% and 20% RS images are randomly selected to train various models. The OAs and standard deviations are shown in Table III, where it can be found that T-CNN outperforms all the comparison networks. Under the training ratio of 10%, T-CNN has 3.1, 1.03, 1.22, 0.36, 0.46, 0.95, 2.34, 0.75, and 0.08 percentage points' increases over fine-tuned VGGNet-16, D-CNN, Inception-v3-CapsNet, SF-CNN, RAN, SCCov, CNN-MS2AP, GLANet, and MF2Net, respectively. With the training proportion of 20%, compared to the other methods, the advancements obtained by T-CNN are 2.69 (fine-tuned VGGNet-16), 1.16 (D-CNN), 0.45 (Inception-v3-CapsNet), 0.5 (SF-CNN), 1.65 (RAN), 0.95 (SCCov), 2.07 (CNN-MS2AP), 1.55 (GLANet), and 0.32 (MF2Net) percentage points. CMs of T-CNN on the NWPU-45 dataset are shown in Figs. 13 and 14. After observing the matrices, it can be concluded that T-CNN is effective for most scene categories. The number of categories whose accuracies are higher than 90% is 27 of the 45 categories. The accuracy of T-CNN is greater than 85% for 40 of the 45 categories. These promising results indicate that the proposed method is effective for the RS scene classification task, even if the dataset is complicated.

4) Comparisons With Other Transfer Learning-Based Methods: The proposed T-CNN is expected to provide an adaptive strategy based on transfer learning for RS scene classification. To show the advantages of the proposed T-CNN, we compare the proposed model with several methods based on transfer learning for RS scene classification. In [2], [20], [22], and [49], the fine-tuning technique is used to pretrain models on the source domain. They both use the architectures of pretrained CNNs as feature descriptors and transfer the extracted features into the RS scene classification task. This kind of transfer learning method (fine-tune) is usually applied between homogeneous networks, which may not be the optimal choice when the source and target tasks are discrepant, especially for natural images and RS scenes. T-CNN utilizes a heterogeneous network based on transfer learning for RS scene classification, which considers the differences between natural image classification and RS scene classification. The experimental results show that the proposed T-CNN outperforms the methods mentioned above.

In general, the reasons for the superior behavior of the proposed method are threefold. First, due to the proposed preprocessing module of T-CNN-AT, the appropriate transformation of RS images can be explored as suitable for the RS scene classification task. Meanwhile, T-CNN-AT resizes the original RS images to the appropriate size of the transfer model adaptively without losing abundant information, unlike the classical interpolation and random cropping operations. Second, because of the introduced transfer loss, which is weighted by metanetworks, the T-CNN-AM not only can learn which features are vital to transfer from the pretrained CNN model but is also capable of deciding which layers are proper to transfer. In this way, the RS scene classification model always achieves better performance than the common transfer networks, which transfer features manually. Third, with the help of the T-CNN-AL, the connections between similar categories are taken into account during the training phase, which is reflected in the adaptive labels based on the model predictions. All of these are conducive to highlighting the target-level information further and making the proposed model an improvement over state-of-the-art models.

5) Analysis of Parameters: The parameter $\rho^{a,b}$ weighs the amount of transfer between pairs of layers from the pretrained CNN model to the RS scene classification model, which plays

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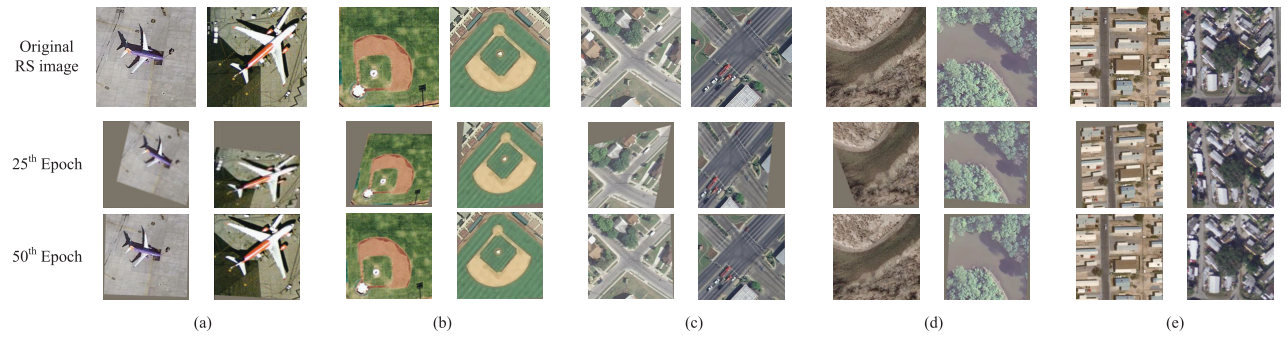


Fig. 16. Several experimental results of T-CNN-AT that are randomly selected in the 21th epoch and the 41th epoch. The categories of the selected figures are (a) airplane, (b) baseball diamond, (c) intersection, (d) river, and (e) mobile home park.

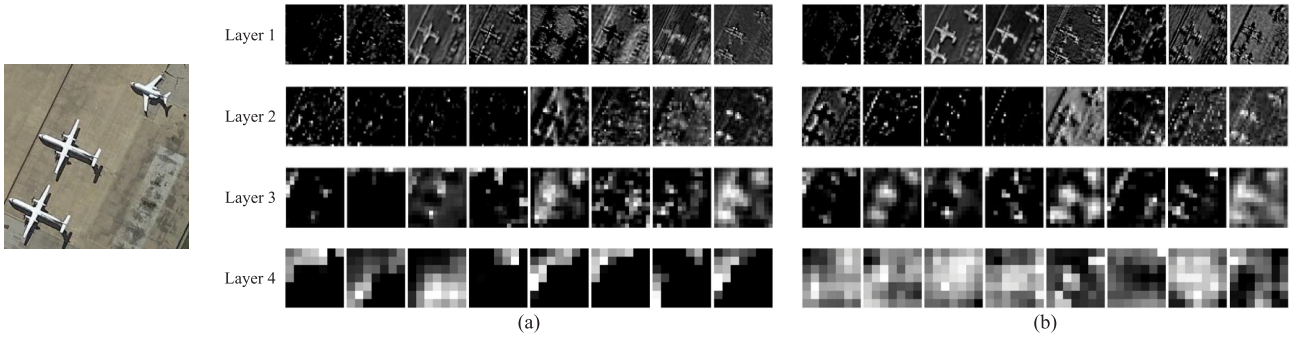


Fig. 17. Feature map visualization of the pretrained ResNet-18 and T-CNN-AM on the NWPU-45 dataset. An "Airplane" RS image is randomly selected during testing and used to generate feature maps. The feature maps are selected from different layers of the RS classification models. (a) Pretrained ResNet-18. The weights are transferred directly. (b) T-CNN-AM. The classification model is still ResNet-18, but the weights are automatically transferred with the proposed T-CNN-AM method.

an important role in T-CNN. Fig. 15 shows the mean values and standard deviations of transfer parameter $\rho^{a,b}$ between the layers of the pretrained CNN model and the RS scene classification model on three benchmark datasets. As shown in the figure, the higher values of $\rho^{a,b}$ are $\rho^{1,3} = 1.63$, $\rho^{1,2} = 1.47$, $\rho^{2,3} = 1.18$, and $\rho^{4,2} = 2.09$. The weights of some compared pairs are lower than 1.0 (e.g., $\rho^{1,1}$, $\rho^{2,2}$, $\rho^{3,1}$, and $\rho^{4,3}$), which are not trivial to find by fine-tuning manually. These results demonstrate that T-CNN-AM is beneficial for learning which layer to transfer. Furthermore, during the training phase, the values of transfer parameter $\rho^{a,b}$ are continuously adjusted effectively. One conspicuous phenomenon that can be found is that values of $\rho^{a,b}$ from the features of the pretrained CNN model $\mathcal{S}^1(\boldsymbol{x})$ have relatively lower variance than those of $\mathcal{S}^3(\boldsymbol{x})$ and $\mathcal{S}^4(\boldsymbol{x})$, which is because lower level features are more task-agnostic, while higher level features are more task-specific. The mean values of $\rho^{a,b}$ from lower layers have smaller fluctuation than that from higher layers. However, the different weight values learned from the pretrained CNN model prove that T-CNN-AM is able to transfer knowledge according to its importance to the RS classification model. These results demonstrate that the designed metanetworks adjust the amounts of transfer for different samples considering the relationships between the RS scene classification task and the abstract levels of features.

6) Visualization: To give a more intuitive performance of T-CNN, we visualize the results of three essential components of T-CNN, respectively. In Fig. 16, we randomly select the results of T-CNN-AT in the 21th epoch and the 41th epoch, and the experiment is conducted on the NWPU-45 dataset. During the training process, T-CNN-AT is able to transform the original RS image from the perspectives of angles and scales. As the training continues, T-CNN-AT can preserve most information from the original RS images, and it is able to learn a certain transformation, which changes the size of raw images into the required size of the pretrained CNN model and the RS classification model.

To intuitively interpret the proposed transferring method, we simply feed the resized image into the pretrained ResNet-18 and T-CNN-AM to compare the features between T-CNN-AM and the directly transferred model, and the feature maps of two different models (directly pretrained ResNet-18 and T-CNN-AM) are visualized in Fig. 17. The experiment is run on the NWPU-45 dataset with 10% training samples. In Fig. 17, compared to the feature maps of the pretrained ResNet-18, T-CNN-AM can obtain more abundant fine-grained features by metanetworks, which is able to help the network to learn distinct characteristics of each category. For example, some features of both networks have attention on the airplanes. However, the features captured by T-CNN-AM are clearer than that produced by the pretrained ResNet-18. In addition to airplanes, some patches of T-CNN-AM focus on the runway, which is an important character to classify the "Airplane" scene. Moreover, RS images have wide coverage and large resolution, and extracting local features may help the network to focus on confusing information. In Fig. 17,

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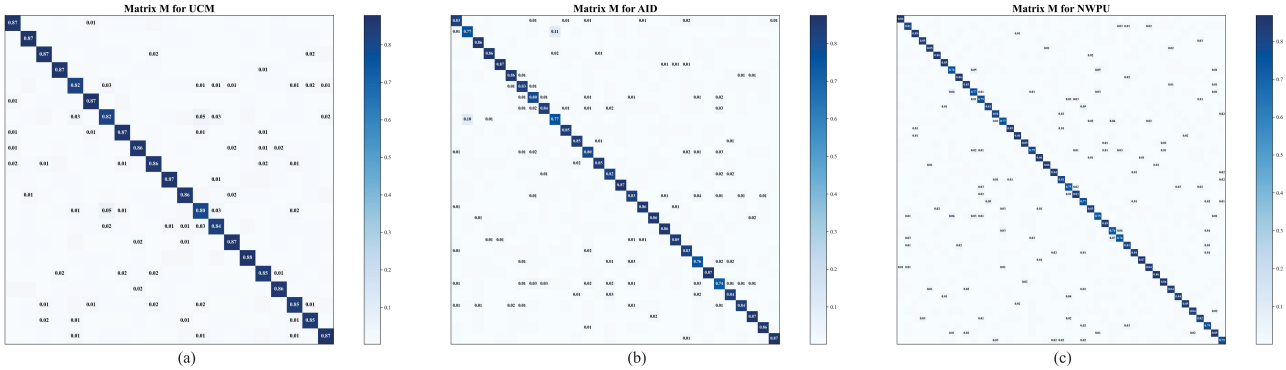


Fig. 18. Learned matrices M on the three datasets. (a) UCM. (b) AID. (c) NWPU-45.

it is apparent that the attention region in some features of T-CNN-AM is more remarkable than that of the pretrained ResNet-18. Furthermore, it can be seen that the feature maps of high levels in T-CNN-AT include more information than that in the pretrained CNN. It is because T-CNN-AM transfers much more knowledge from the features of the pretrained CNN model to the high levels of the classification model, which is helpful for improving the performance of the RS classification task.

To demonstrate the effectiveness of T-CNN-AL, the learned matrix M for each RS dataset is shown in Fig. 18. For a certain dataset, the size of the learned matrix M is $\mathcal{K} \times \mathcal{K}$, where $\mathcal{K}$ is the categories of the dataset. It is worth noting that the matrices in Fig. 18 show the updated label, which are the results of (11). At the end of the current training epoch, the cumulative M should be normalized column by column by (11). From the observation of Fig. 18, the diagonal elements of final learned matrices M maintain the highest value in terms of each column, which indicates the correctness of soft labels. This is because we use a uniform distribution as the soft label in the first epoch of training, which is equivalent to vanilla label smoothing. In the entire training process afterward, we only accumulate correct predictions, which guarantees the correctness of the generated soft labels. Moreover, although the model reaches convergence at the end of the training phase, the soft labels of nondiagonal elements still show the relationship between the target categories and nontarget categories to some extent, which has played an important role in the stage of the model prediction.

7) Flexibility of the Adaptive Modules: To further verify the effectiveness of our proposed T-CNN, we have conducted relevant experiments to show the flexibility of T-CNN. We have implemented a state-of-the-art RS scene classification method called LML [69], which is able to capture the global and local features with mutual learning. The experimental settings are consistent with the original paper except for the batch size (owing to the limitations of GPU memory), which is set to 16 in this part. The experimental results are shown in Fig. 19. From Fig. 19, it can be seen that LML achieves good performance on both datasets. On the basis of LML, we have added two adaptive modules (T-CNN-AT and T-CNN-AL), respectively. The experimental results indicate that our adaptive modules can further enhance the classification performance of RS scenes, which verifies the effectiveness of the proposed T-CNN.

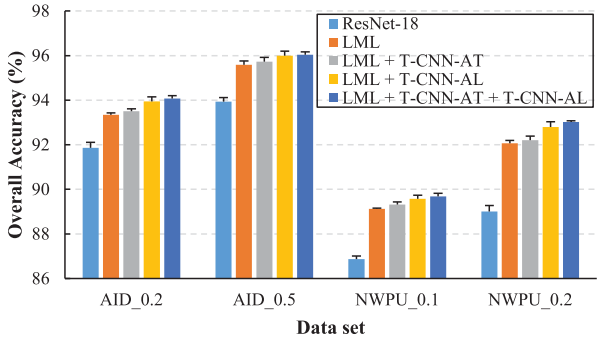


Fig. 19. Experimental results counted on the AID and NWPU-45 datasets with different ratios of training data.

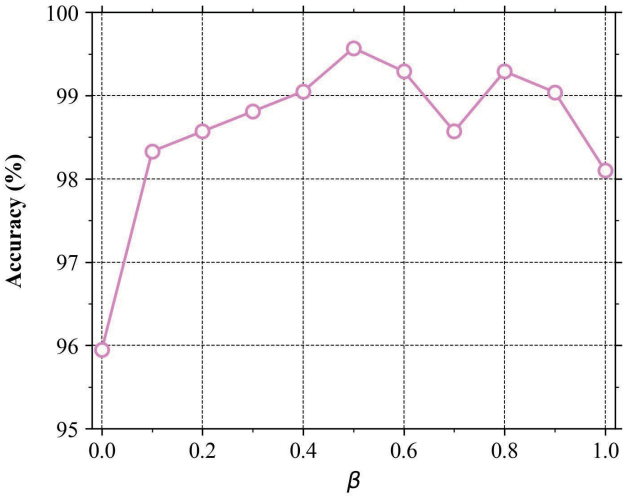


Fig. 20. Impact of hyperparameter β.

8) Hyperparameters: Analyzing the impact of hyperparameters also yields significant findings. The bilevel scheme is used to train T-CNN-AM, which is improved based on [70]. Hyperparameter α (referred to as λ in [70]) in (7) can balance the transfer loss and the original loss (e.g., cross-entropy loss). [70] gradually anneals λ during the training with a linear decay, which allows easier examples to initially have a stronger effect, but progressively decreasing their importance as λ decays. However, in T-CNN-AM, α is just a balanced parameter between two mentioned losses; the key point should

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TABLE IV
ABLATION EXPERIMENTAL RESULTS (VALUES ± STANDARD DEVIATIONS) COUNTED ON DIFFERENT RS IMAGES DATASETS WITH DIFFERENT RATIOS OF TRAINING DATA

Methods				OA/Data Sets (Training Ratio)				
ResNet-18	T-CNN-AT	T-CNN-AM	T-CNN-AL	UCM (80%)	AID (20%)	AID (50%)	NWPU (10%)	NWPU (20%)
✓				97.92 ± 0.22	92.62 ± 0.15	94.35 ± 0.34	87.79 ± 0.40	90.69 ± 0.12
✓	T-CNN-AT-ST			98.10 ± 0.24	92.66 ± 0.08	94.44 ± 0.21	87.86 ± 0.22	90.75 ± 0.19
✓	✓			98.49 ± 0.14	92.85 ± 0.09	94.71 ± 0.11	88.02 ± 0.09	90.96 ± 0.14
✓		✓		98.71 ± 0.13	93.19 ± 0.12	95.20 ± 0.20	88.54 ± 0.08	91.81 ± 0.19
✓			✓	98.57 ± 0.24	93.28 ± 0.16	95.02 ± 0.21	88.73 ± 0.16	91.57 ± 0.19
✓	✓	✓		99.14 ± 0.13	94.11 ± 0.22	95.71 ± 0.14	89.44 ± 0.22	92.61 ± 0.08
✓	✓		✓	98.97 ± 0.13	94.05 ± 0.20	96.05 ± 0.23	89.73 ± 0.19	92.53 ± 0.15
✓		✓	✓	99.24 ± 0.11	94.25 ± 0.15	96.38 ± 0.22	89.83 ± 0.15	92.85 ± 0.11
✓	✓	✓	✓	**99.33 ± 0.11**	**94.55 ± 0.27**	**96.72 ± 0.23**	**90.25 ± 0.14**	**93.05 ± 0.12**

The entry with the highest values is bold-faced.

focus on the adjustments of the values of transfer parameter $\rho^{a,b}$, which are weighted in the transfer loss. Hence, we just need to keep the fixed value of α, which is set to 0.5 in this article. The other hyperparameter is β, as shown in (10). Unlike previous experiments that set β to 0.5 directly, the values of β vary from 0.1 to 1.0 with the interval of 0.1. Fig. 20 shows the experimental results on the UCM dataset. It can be seen that T-CNN obtains the highest accuracy of classification when β is set to 0.5.

In addition, the accuracy drops sharply when $\beta = 0$, and we observe that T-CNN is difficult to converge under this condition. The accuracy of classification persistently increases when β changes from 0.1 to 0.5. It reveals that the LS labels still have a positive impact on the proposed model by providing the correct category information. Meanwhile, when $\beta = 1$, which means that the model only uses LS regularization into T-CNN, classification accuracy is lower than other values of β (except for $\beta = 0$). This suggests that T-CNN-AL is beneficial for improving the performance of the RS scene classification model.

V. ABLATION STUDY

As mentioned in Section III, the proposed strategy mainly contains T-CNN-AT, T-CNN-AM, and T-CNN-AL. To study their influence on the proposed strategy, in this section, we perform related ablation experiments.

A. Comparisons With Different Models

The experimental settings are consistent with the contents of Section IV-C. The results of three RS image datasets are shown in Table IV. The networks, which exclude the T-CNN-AT, all preprocess the original RS images with bilinear interpolation. The eight networks described in Table IV are conducted on three datasets in order to study the behavior of different models. From these observations, several conclusions are made and summarized in the following. The performance of the baseline model (ResNet-18) is the weakest among the eight networks due to the knowledge deficiency and, to a certain extent, the overfitting phenomenon. With the aid of the T-CNN-AM, the RS scene classification model transfers the prior knowledge from the pretrained CNN model adaptively, manifesting the better behavior of the third network than that of the first network. Moreover, it can be seen that the performance of the second network is improved when the T-CNN-AT is embedded in the preprocessing phase. It suggests that T-CNN-AT exploits the initial information of the RS images more sufficiently than methods that interpolate the images directly. Furthermore, after adding the T-CNN-AL strategy, the performances of the fourth network and the seventh network are further enhanced for different datasets. It shows that this kind of strategy, which updates the supervised labels dynamically, plays a fairly positive role in the RS scene classification task.

TABLE V
COMPARISON OF COMPUTATIONAL COSTS AND TIMES ON THE UCM DATASET

Methods	Parameters (M)	FLOPs (G)	Time (min)
ResNet-18	11.19	1.82	15.17
ResNet-18 + T-CNN-AM	12.59	1.91	117.53
ResNet-18 + T-CNN-AT	14.5	4.31	148.9
T-CNN	15.9	4.67	153.28

B. Effectiveness of Rotation

Many RS scene classification methods pay attention to rotation invariance features of RS scenes. It is worth noting that T-CNN-AT also takes rotation into consideration. However, the learnable parametric matrix, $\mathbf{\Theta}$, tends to consider three transformations simultaneously, including rotation degrees, scale sizes, and translation distances. Consequently, we design another ablation experiment to verify the effectiveness of rotation.

According to (2), matrix $\mathbf{\Theta}$ consists of six parameters, including θ_{11}, θ_{12}, θ_{13}, θ_{21}, θ_{22}, and θ_{23}, where θ_{11}, θ_{12}, θ_{21}, and θ_{22} are able to rotate and scale the input image, and θ_{13} and θ_{23} are capable of translating the image. In fact, it is difficult to separate rotation and scaling transformations according to θ_{11}, θ_{12}, θ_{21}, and θ_{22}. In contrast, we design T-CNN-AT-ST that only considers scaling and translation transformations. The T-CNN-AT-ST is easy to be implemented, which only needs to let $\theta_{11} = \theta_{22}$ and $\theta_{12} = \theta_{21} = 0$. In other words, T-CNN-AT-ST just needs to learn three

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parameters of matrix $\boldsymbol{\Theta}$. Relevant experimental results are shown in Table IV. From Table IV, it is obvious that T-CNN-AT achieves the best performance in terms of ResNet-18, T-CNN-AT-ST, and T-CNN-AT. T-CNN-AT-ST, which does not perform a rotation transformation, just slightly outperforms ResNet-18. This result testifies the significance of the rotation transformation for RS scene classification.

C. Computational Complexity Analysis

Compared to the baseline model (ResNet-18), T-CNN has more computation cost during the inference time. T-CNN-AM and T-CNN-AT occupy the most computation cost. From Tables IV and V, it can be seen that T-CNN-AM is able to gain 0.5%∼1.2% OA improvement with almost no extra computation costs. However, the training time of T-CNN-AM is more than that of ResNet-18 due to the alternative training scheme of metanetworks and the RS classification model. As for T-CNN-AT, more convolutional layers are introduced, which leads to more convolution operations. A fully connected operation also increases the computation cost. However, in T-CNN-AM, the input size of the pretrained CNN model and RS classification model is always fixed (e.g., 224 × 224 pixels), which is inconsistent with the input size of RS images. T-CNN-AT provides a viable strategy to solve this mismatching problem and preserves the key information of the RS images as much as possible by learning effective transformations of RS images. Moreover, in Table IV, T-CNN-AT can also earn accuracy improvement by a big margin (about 1% in terms of OA in some cases) even though some extra inference computation is introduced.

VI. Conclusion

In this article, an adaptive learning strategy called T-CNN has been proposed for transferring useful knowledge from the source domain to the RS scene classification task automatically. The proposed T-CNN solved the problematic issues of previous transfer learning-based models from three perspectives: input, model, and label. Specifically, through T-CNN-AT, the proposed strategy first transformed the original RS images into a size required by the input of the pretrained CNN model. T-CNN-AT not only preserves key information in RS images but is also able to learn the rotation-invariant of class-specific ground objects. Then, T-CNN-AM based on metanetworks was utilized to autonomously determine which information should be transferred from the pretrained CNN model to the RS scene classification model and which layers were connected appropriately. Furthermore, by using T-CNN-AL, the reasonable relationships of various categories within the RS scenes, which were embodied in model predictions, were taken into consideration. The experiments on three available benchmark datasets demonstrated the superiority of the proposed strategy compared to other state-of-the-art RS scene classification models. We are hoping that the adaptive learning strategy proposed here will inspire further innovation and promote the development of transfer learning in the domain of RS scene classification. We intend to investigate approaches based on transfer learning to further improve classification performance in the future.

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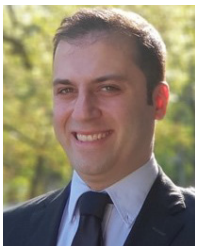
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